

Geo Employee Tracker — Complete QA & Bug Audit Report

Application: `eb_geoemployee_tracker_mobile` **Date:** 14 August 2026 **Audit type:** End-to-end static code review (architecture, APIs, navigation, security, Android/iOS). Not a device lab test. **Stack:** React Native CLI **0.86.0**, React **19.2.3**, Redux Toolkit, Axios (default timeout 15s), Odoo REST. Not Expo. **Scope:** Entire mobile app. No application code was modified during this audit. **Secrets:** Firebase / Google Maps keys, tokens, and passwords are **not** reproduced in this report.

1. Cover / architecture snapshot

Auth is Redux Toolkit plus Axios against a user-entered Odoo HTTPS base URL. Tokens live in AsyncStorage. There is no auth-gated navigator: every screen is on one native stack.

Main flow: Fetch URL → Login → Home tools (Geo / Attendance / Time Off) → field check-in, default-route GPS session, task routing + Google Maps WebView, geo history, HR leave.

Session sync is TTL-deduped (15 seconds). Token refresh is single-flight on HTTP 401. Several screens cancel GETs with `AbortController`; several high-traffic screens do not.

2. Executive summary

Area	Count (approx.)
Total issues	62
Critical	6
High	18
Medium	24
Low	14
API	22
UI/UX	12
Performance	8
Security	8
Navigation	5
Android	7
iOS	5

How to read findings

- **Confirmed** = the code path exists today.
- **Potential** = it can happen under timing, bad API data, or device conditions.

3. Issue table

3.1 Critical

ID	Category	File	Issue	Why it happens	User impact	Recommended fix
C1	Auth / Nav	<code>App.tsx</code> , <code>AppNavigator.tsx</code>	401 refresh failure logs the user out in Redux/storage but does not reset navigation.	<code>onAuthFailure</code> calls <code>logout()</code> / <code>clearSession()</code> ; the stack is not auth-aware. Settings logout <i>*does*</i> <code>reset</code> .	User stays on Home/tasks with a dead session.	<code>navigationRef.reset</code> to Fetch URL/Login. Gate stack on <code>accessToken</code> .
C2	Location	<code>locationTrackingService.ts</code> , <code>AndroidManifest</code> , <code>Info.plist</code>	Tracking is a JS <code>setInterval</code> with no Android foreground service, no <code>ACCESS_BACKGROUND_LOCATION</code> , iOS background mode is only <code>remote-notification</code> .	Interval pauses when JS is suspended.	Field path stops in background . Core product gap.	Native background location + foreground service + Always permission; queue uploads.
C3	Security	<code>sessionStorage.ts</code> , <code>AndroidManifest</code>	Access + refresh tokens in AsyncStorage ; <code>android:allowBackup="true"</code> .	Backup/root can copy <code>@geo_employee_tracker/session</code> .	Account takeover.	Encrypted storage (Keychain/Keystore); <code>allowBackup="false"</code> .
C4	Privacy	<code>locationTrackingService.ts</code>	Production <code>console.log</code> of <code>lat/lng</code> and <code>point_uid</code> .	Logs not wrapped in <code>__DEV__</code> .	PII in logcat / crash tools.	Log only in <code>__DEV__</code> ; never log coordinates.
C5	Android	<code>AndroidManifest.xml</code>	<code>android:icon="@mipmap/ic_launcher"</code> has a space .	Invalid resource name.	Icon/build failure or launcher	Use <code>@mipmap/ic_launcher</code> .

					crash.	
C6	Maps / Security	TaskMapScreen.tsx , taskMapHtml.ts , GeoHistoryMapScreen.tsx	Company Google Maps JS key injected into WebView HTML.	fetchFieldMapConfig returns the key; HTML loads Maps JS with it.	Key extraction, billing abuse if unrestricted.	Restrict key; prefer native map SDK; keep Directions on server.

3.2 High

ID	Category	File	Issue	Why it happens	User impact	Recommended fix
H1	API / UX	TimeOffCalendarScreen.tsx	List uses default limit: 20 , no pagination.	fetchHrLeaves defaults to 20; screen never loads more.	Older requests missing.	Paginate or load all pages.
H2	Race	TimeOffRequestScreen.tsx	setSubmitting(true) runs after validation.	Two fast taps both see submitting === false.	Duplicate leave records.	Lock with a ref immediately on first tap.
H3	API duplicate	TaskMapScreen.tsx	Every focus refetches task detail, map-config, and directions.	useFocusEffect → bootstrap; HTML cleared.	Latency, quota, flicker.	Bootstrap once per taskId; cache config.
H4	Performance	TaskMapScreen.tsx	watchPosition with distanceFilter: 0, 1s / 500ms.	Watch starts on every focus.	Battery drain, heat.	Filter 5-15m; interval 3-5s.
H5	API	TaskRoutingScreen.tsx	No AbortSignal . Refresh uses loadTasks(() => false).	Chip/date/page/focus all call GET; old request continues.	Duplicate GETs; setState after unmount.	AbortController; cancel on unmount.
H6	API duplicate	Time Off Calendar + Request	Calendar loads leave + types on every focus; Request loads types again + up to 200 leaves.	No shared cache.	Slow Time Off flow.	In-memory TTL cache.
H7	API duplicate	HomeScreen.tsx + TaskMapScreen.tsx	Home fetches task detail to navigate; Map fetches again.	Two GET /field/tasks/{id} in one flow.	Delay opening the map.	Pass detail in params or short cache.
H8	Prod / Privacy	apiClient.ts	[TimeOff] logs run in production .	No __DEV__ guard.	PII in device logs.	Guard or delete.
H9	Error UX	apiClient.ts	Unknown 500/403/422 fall through to Axios/data.error.	Only some codes mapped.	Users see Request failed with status code 500.	Map all 4xx/5xx to safe copy.
H10	Tracking	locationTrackingService.ts	Failed uploads dropped; is_offline_sync unused.	Catch logs and returns; no queue.	Gaps in the trail offline.	Persist a point queue.
H11	Auth UX	Screens generally	After C1, screens show "Session expired" in place .	No global redirect.	Stuck authenticated UI.	Same as C1.
H12	Time	CheckInScreen.tsx	Naive datetimes get a Z suffix .	Treated as UTC.	Wrong worked hours vs local TZ.	Parse as API actually sends time.
H13	Data	auth employee	Time Off fails if employee.id missing.	MobileEmployee.id?: number.	Cannot load/submit leave.	Require id at login.
H14	Security	android/app/google-services.json	Firebase client key is committed .	Standard RN Firebase file in git.	Abuse if API restrictions are loose.	Restrict by app id/SHA-1; rotate if leaked.
H15	Timeout	apiClient.ts	Default 15s ; uploads 60s; stop images 120s ; no retry except 401.	Axios default.	Slow Odoo hits timeout.	15s GET, 30s mutations, 60s media; one network retry.
H16	Leak	TaskMapScreen.tsx	Detail/config/directions omit signal .	Cancel flag ignores in-flight HTTP.	Calls continue after leaving the map.	Pass AbortSignal; abort on blur.
H17	Interval	locationTrackingService.ts	Backend can set 1s GPS upload.	uploadInFlight skips overlaps.	Battery + missed points.	Client floor 10-15s.
H18	UX dead	HomeScreen.tsx	Get Help onPress={() => {}}.	Stub.	Looks broken.	Wire help URL or hide.

3.3 Medium

ID	Issue	Plain explanation
M1	Home focus session sync	After 15s TTL, every Home focus attempts GET /field/session.
M2	Resume prompt once	Tapping Later never asks again until Home remounts.
M3	Check-in double session GET	Focus + submit both call sync; TTL usually collapses them.
M4	Task Routing GPS 4s race	Losing Promise.race does not cancel getCurrentCoordinates; dialog can pop late.
M5	History day max 50 sessions	limit: 50, offset: 0 only. Extra sessions that day hidden.
M6	Occupied dates from first 200 leaves	User might book a day that already has leave.
M7	Login email	Only checks empty, not format; rare double-tap before loginLoading.
M8	Fetch URL search	Same one-frame double search.

M9	Profile photo as huge base64	Memory + large PATCH.
M10	Check-in clock every 1s	Whole screen re-renders.
M11	Notification channel deleted then recreated	User mute/importance reset.
M12	Foreground push only for no-movement	Other FCM types silent in foreground.
M13	iOS When In Use only	Same background gap as C2 on iOS.
M14	No Android foreground location service	Background tracking cannot ship without this.
M15	Invalid Maps key still loads HTML	Blank map possible.
M16	Task Map lat/ing not validated	NaN map / HTML crash if params bad.
M17	Logout revoke 15s × 2	Best-effort revoke can hang logout.
M18	Splash blocks forever	GPS/camera/network alerts are non-cancelable.
M19	Access gate on all post-splash screens	Cannot view Time Off without camera.
M20	View Details note effect	Params + local note can fight / extra renders.
M21	Task Capture refetch on every focus	Extra GET after camera.
M22	Checklist refetch on every focus	Extra GET after map.
M23	Silent session GET failure	Stale checked-in if server down; no toast.
M24	Time Off kebab stub	More-options does nothing.

3.4 Low

ID	Issue	Recommended fix
L1	Unused packages: MapLibre, <code>react-native-maps</code> , Nitro Image, <code>@react-native/new-app-screen</code>	Remove unused packages.
L2	Two vector-icon libraries	Keep one.
L3	ProGuard/R8 off in release	Enable with keep rules.
L4	Theme follows OS only	Optional in-app setting.
L5	Terms URL <code>Linking.openURL</code> without try/catch	Catch + toast.
L6	Tablet <code>contentMaxWidth: 480</code>	Widen if tablets supported.
L7	Portrait lock both platforms	Intentional; none.
L8	Unguarded TaskMap/TaskRouting logs	Guard with <code>_DEV</code> .
L9	<code>ApiClient.ts</code> ~2400 lines	Split by domain (optional).
L10	Fetch URL rejects ports / localhost	Allow port in debug builds.
L11	No app deep-link intent filters	Add if notifications should open a task.
L12	iOS ATS <code>NSAllowsArbitraryLoads</code> false	Keep HTTPS-only (good).
L13	<code>usesCleartextTraffic</code> placeholder	False in release.
L14	Almost no tests	Add API/auth tests.

4. API call summary

Timeouts: Axios **15s** default. Longer: check-in/out, start/end session, add-stop, location upload, leave attach, selfie (**60s**); stop images (**120s**). **Retry:** HTTP **401 refresh only** (single-flight). **Cancellation:** mixed.

API	Method	Called from	Timeout	Retry	Duplicate risk	Recommendation
<code>/api/v1/company/search</code>	GET	Fetch URL	15s	No	Low (double-tap 1 frame)	Submitting ref
<code>/api/v1/auth/login</code>	POST	Login	15s	No	Low	Disable immediately
<code>/api/v1/authentication/token</code>	POST	Splash + Axios 401 interceptor	15s	No	Low if splash+API overlap	Share single-flight with splash
<code>/api/v1/authentication/revoke</code>	POST	Logout (access + refresh)	15s	No	None	Short timeout (3-5s)
<code>/api/v1/me</code>	GET	Home/Profile if no employee	15s	401	Low	OK
<code>/api/v1/me</code>	PATCH	Profile save	15s	401	Low	Already <code>busy</code>
<code>/api/v1/field/session</code>	GET	Splash, Home focus, CheckIn, DefaultRoute, login sync	15s	401	Medium after TTL	Keep TTL; toast on failure
<code>/api/v1/field/check-in /check-out</code>	POST	CheckIn	60s	401	Low	OK (<code>submittingRef</code>)
<code>/api/v1/field/start-session /end-session</code>	POST	Default Route	60s	401	Low	OK
<code>/api/v1/field/location</code>	POST	Interval tracker	60s	401	High frequency	Min 10-15s; queue

/api/v1/field/device-token	POST/DELETE	Login/splash/logout	15s	No	Low	OK additive
/api/v1/field/tasks	GET	Task Routing focus/chip/date/page/refresh	15s	401	High (no abort)	Abort + debounce chip
/api/v1/field/tasks/{id}	GET	Home resume, TaskMap, Capture, Checklist, History	15s	401	High	Cache; don't refetch every focus
/api/v1/field/map-config	GET	TaskMap + GeoHistoryMap	15s	401	High on TaskMap focus	Cache for session
/api/v1/field/directions	POST	TaskMap route + watch reroute	15s	401	High	Abort; keep throttle
/api/v1/field/attendance/calendar	GET	Attendance month; TimeOff date modal	15s	401	Low	OK
/api/v1/field/history	GET	Geo History month	15s	401	Low	OK
/api/v1/field/history/years	GET	Geo History mount	15s	401	Low	OK
/api/v1/field/history/sessions	GET	Day screen	15s	401	Low	Paginate if total > 50
/api/v1/field/sessions/{id}	GET	Session screen	15s	401	Low	OK
/api/v1/field/sessions/{id}/route	GET	History map	15s	401	Low	Parallel OK
/api/v1/records/hr.leave	GET	Calendar focus; Request occupancy	30s	401	High	Cache; paginate
/api/v1/records/hr.leave.type	GET	Calendar + Request	30s	401	High	Cache
/api/v1/records/hr.leave	POST	Request submit	15s	401	High	Immediate lock
hr.leave/{id}/attachments	POST	After create	60s	401	Low	Retry attach without second leave
Task start/arrive/complete/cancel/pause	POST	Map/Capture	15s	401	Medium	Keep refs
Stop note/selfie/images	PATCH/POST	Capture	60-120s	401	Medium	Cancel on leave

Over-called (do not remove the API; reduce frequency): GET /field/session on every Home focus after TTL; GET /field/tasks/{id} + map-config + directions on every TaskMap focus; hr.leave + hr.leave.type on every Time Off Calendar focus and again on Request; Task Routing GET on every focus when returning from a task.

Required: login, session sync after check-in, location upload while tracking, task mutations, leave create.

5. Screen-by-screen summary

Screen	API issues	UI issues	Navigation	Performance	State	Severity
Splash	Session refresh + field/session + push	Blocks on GPS/camera/network	replace Home/Fetch URL	Sequential waits	Cancel on unmount OK	Medium
Fetch URL	Search double-tap potential	KeyboardAvoiding OK	From splash via replace	—	URL persisted	Low
Login	Weak email validation	Terms + password toggle OK	reset Home	—	Redux loading	Medium
Home	Session sync on focus; resume task-detail; Get Help stub	Check In/Out label OK	Resume → Map/Capture	—	resumePromptedRef	High
Settings	Logout revoke	Logout color hardcoded	reset Fetch URL (good)	Revoke wait	—	Low
Profile	Conditional GET me; large avatar PATCH	Edit/read modes	Back OK	Base64 memory	Form vs employee	Medium
Check-In	Session GET + check-in/out	1s clock rerender	Selfie round-trip	Timer	TZ Z bug	High
Selfie camera	None	VisionCamera focus	returnTo checkIn/visitStop	Camera lifecycle	beforeRemove while capturing	Low
Geo Management	None	Rows OK	To Default/Task Routing	—	—	Low
Default Route	Session GET; start/end	Hydrate spinner	Check-in CTA	—	submittingRef OK	Medium
Task Routing	Tasks GET no abort; GPS race	Pagination UI	To map/checklist	GPS 4s	Chip/page refetch	High
What Needs To Be Done	Detail every focus	Checklist	Back	—	Local progress merge	Medium
Task Map	Detail+config+directions every focus; 1Hz GPS	WebView map; banners	Back; start/arrive	Battery	Many refs	Critical/High
Task Capture	Detail every focus; selfie/complete	Requirements gating	Pop to routing	Image upload	Dual focus work	Medium
View Details	add-stop	Notes/photo	Default-route stop	—	Param sync	Medium
Geo History	years + month history	Month pills	To day	—	Abort OK	Low
Geo History Day	Sessions cap 50	Cards	To session	—	—	Medium
Geo History Session	Session GET	—	To map/task	—	Abort OK	Low

Geo History Map	map-config + route	WebView	Back	HTML map	Abort OK	Medium
Geo History Task Detail	Task + stop images	Image viewer	Back	Base64 images	Abort OK	Medium
Attendance Calendar	Month GET	Calendar/list	—	Abort OK	Month reload	Low
Time Off Calendar	Leaves 20 + types every focus; kebab stub	FAB; chips	To Request	—	Abort OK	High
Time Off Request	Types + 200 leaves + holiday calendar; double submit	Date modal; attachments	goBack	Logs PII	Occupancy incomplete	High

6. Bug explanations (plain language)

6.1 Critical (fix first)

C1 — Session dies, but the app still looks logged in

Where: `App.tsx` (`onAuthFailure`) + `AppNavigator.tsx`

If the access token expires and refresh fails, Redux and storage are cleared, but the stack is **not** sent back to Fetch URL / Login. Settings logout *does* reset navigation. Silent 401 logout does not.

User sees: Home, maps, time off still on screen. Buttons fail or show “session expired” with no way out except force-close.

Fix: On auth failure, `navigationRef.reset` to Fetch URL (or Login). Optionally hide the logged-in stack when `accessToken` is null.

C2 — GPS tracking stops when the app is in the background

Where: `locationTrackingService.ts` (`setInterval`), `AndroidManifest`, iOS `Info.plist`

Uploads run on a JS timer. There is no Android foreground service, no `ACCESS_BACKGROUND_LOCATION`, and iOS only has `remote-notification` (not location) background mode.

User sees: Trail gaps when they lock the phone or switch apps. No-movement alerts and history become wrong. This is the core of a field tracker.

Fix: Native background location + notification; queue points when offline; do not rely on JS `setInterval` alone. This is a **large** native change (later epic).

C3 — Tokens stored insecurely + Android backup

Where: `sessionStorage.ts` (`AsyncStorage`), `AndroidManifest.xml` `allowBackup="true"`

Access + refresh tokens sit in plaintext `AsyncStorage` and can be included in device backup.

Impact: Stolen session / account takeover on a lost or backed-up phone.

Fix: Keychain (iOS) / Keystore (Android). Set `allowBackup="false"`.

C4 — Live GPS logged in production

Where: `locationTrackingService.ts` → `uploadCurrentPoint`

`console.log` prints latitude, longitude, `point_uuid` **without** `__DEV__`.

Impact: Coordinates in Logcat / crash tools (privacy).

Fix: Remove or wrap in `__DEV__`. Never log coordinates.

C5 — Android launcher icon path is broken

Where: `android/app/src/main/AndroidManifest.xml`

`android:icon="@mipmap/ic_launcher"` has a **space** after `mipmap`.

Impact: Bad icon, failed resource compile, or launcher crash.

Fix: `@mipmap/ic_launcher` (no space).

C6 — Google Maps JS API key inside the WebView

Where: `TaskMapScreen.tsx`, `taskMapHtml.ts`, `GeoHistoryMapScreen.tsx`

Backend `map-config` returns the company Google key; the app injects it into HTML. Anyone can extract it from the WebView.

Impact: Billing abuse if the key is not tightly restricted.

Fix: Restrict the key (package name / SHA-1 / HTTP referrer). Prefer a native map SDK. Keep Directions on the server (already proxied). Do not paste the key into chats or this report.

6.2 High

H1 — Time Off list only loads 20 requests

`TimeOffCalendarScreen` → `fetchHrLeaves` default `limit: 20`. No pagination. Older leaves missing; filters look empty.

H2 — Double-tap can create two time-off requests

`handleSubmit` checks `if (submitting)` return first, but `setSubmitting(true)` is **after** validation. Two fast taps both pass. Duplicate leave records.

H3 — Task Map reloads APIs every time you come back

Every focus: task detail + map-config + directions; WebView HTML is cleared. Flicker, delay, extra Google/Odoo quota.

H4 — Task Map GPS watch is too aggressive

`distanceFilter: 0`, `interval 1000ms`, `fastestInterval 500ms`. Battery drain and extra reroute calls.

H5 — Task list requests are not cancelled

Chip/date/page/focus fire `GET /field/tasks` with **no AbortSignal**. Pull-to-refresh uses `loadTasks(() => false)`, so leaving the screen does not cancel. Duplicate GETs; possible state update after unmount.

H6 — Time Off Calendar and Request hit the same APIs twice

Calendar: `hr.leave` + `hr.leave.type` on **every focus**. Request: types again + up to 200 leaves. Slow Time Off flow.

H7 — Opening a task from Home hits task-detail twice

Home already `GET /field/tasks/{id}` to navigate; Map fetches it again.

H8 — Time Off debug logs run in production

`ApiClient.ts` logs `employeeId`, `dates`, `names` with no `__DEV__`.

H9 — Ugly / raw API errors shown to users

Unmapped 500/403/422 become Axios text or raw Odoo `error`. Users see `Request failed with status code 500`.

H10 — Offline GPS points are thrown away

Upload failure is logged and dropped. Types have `is_offline_sync` but it is unused. Holes in the trail with no internet.

H11 — No redirect after interceptor logout

Same as C1 from the user's point of view.

H12 — Check-in times can be wrong (timezone)

Datetimes without `z` get `z` appended, treated as UTC. Wrong clock / worked hours.

H13 — Time Off breaks if `employee.id` is missing

Login employee type has optional `id`. List/submit skip or toast "profile missing".

H14 — Firebase client key is in git

Normal for Firebase, but dangerous if the key has no app/SHA restrictions. Restrict in Google Cloud; rotate if it was shared widely.

H15 — Timeout strategy is uneven

15s default prevents infinite hang; some GETs have no extra timeout; device-token register has no extra timeout; no retry except 401.

H16 — Task Map HTTP continues after you leave

`cancelled` flag ignores in-flight HTTP. `AbortController` on blur needed.

H17 — Server can set GPS upload every 1 second

Client allows 1s; uploads take up to 60s; `uploadInFlight` skips, so points are lost and battery dies. Floor at 10-15 seconds.

H18 — "Get Help" does nothing

`HomeScreen` `onPress={() => {}}`. Open a URL or hide the button.

6.3 Medium

See issue table M1-M24. Highlights:

- **M2:** Resume task "Later" never re-prompts until Home remounts.
- **M4:** Location dialog can appear after the task list already loaded.
- **M5/M6:** Pagination caps hide data (50 sessions / 200 leaves).
- **M19:** GPS + camera required on **all** post-splash screens including Time Off.

- **M23:** Session GET failure silently uses local attendance (stale check-in).
- **M24:** Time Off kebab is a stub.

6.4 Low

Unused map libraries, two icon packages, ProGuard off, almost no tests, huge `ApiClient.ts`, Fetch URL rejects staging URLs with ports, no deep links.

6.5 Auth, timeout, offline, crash (short)

Auth: Tokens in AsyncStorage; Bearer per call (not Axios default header). Refresh is single-flight on 401. Splash refresh is a **second path**. No proactive refresh before expiry (30s skew only at splash). Logout from Settings is complete; interceptor logout is **not**. Push unregister on logout/auth failure.

Timeout / hang: 15s default prevents infinite Axios hang. Splash/GPS waits **can** feel infinite. `slowWaitToast` only tracks **non-GET** after 3.5s. GET screens can spin until 15s then error (good) or until abort.

Offline: NetInfo on splash only. Later screens map Axios no-response to `network_error`. Location points are **not** queued. Attendance sync **silently** uses cache.

Crashes (realistic): invalid map coords; WebView/Maps JS; missing `employee.id`; unhandled `setState` on TaskRouting refresh; VisionCamera if no device; assuming `checkList.todo` arrays (parser throws → error UI, not crash).

7. Fix priority

7.1 Fix immediately (Critical + High)

1. **C1** — Redirect to login when refresh fails (auth-aware navigation).
2. **C2 / H10 / H17 / M13 / M14** — Real background tracking + upload queue + sane min interval (C2 is a large follow-up).
3. **C3** — Secure token storage; disable backup.
4. **C4 / H8 / L8** — Strip production logs (especially GPS and Time Off).
5. **C5** — Fix launcher icon resource name.
6. **C6 / H14** — Restrict Maps/Firebase keys; stop treating JS Maps key as safe in HTML.
7. **H1 / H2 / H6** — Time Off pagination, submit lock, shared leave-type cache.
8. **H3 / H4 / H16** — Task Map: don't rebuild on every focus; throttle GPS; abort HTTP.
9. **H5** — Abort Task Routing GETs; fix refresh cancellation.
10. **H9** — Safe API error copy.
11. **H12** — Check-in time zone.
12. **H18** — Remove or implement Get Help.

7.2 Fix before production (Medium)

M2 resume prompt, M4 GPS race, M5/M6 pagination caps, M7/M8 double-submit, M9 avatar size, M10 clock interval, M11 notification channel, M19 camera gate on Time Off, M21/M22 focus refetch, M23 silent session errors, M24 kebab, login email format, ProGuard (L3) if you ship obfuscation.

7.3 Improvements (Low)

Remove unused MapLibre / RN Maps / Nitro Image / new-app-screen. Split `ApiClient.ts`. Tests for auth interceptor and attendance TTL. Optional in-app theme. Tablet width. Deep links if notifications should open a task.

7.4 Suggested PR split

1. C5, C4/logs, H2, H18, H17, Manifest backup (small, safe)
2. C1 navigation + H9 errors
3. Time Off H1/H6 + kebab
4. Task Routing H5 + Task Map H3/H4/H16 + Home H7
5. Check-in timezone + access gate + session toast
6. Later epic: **C2 background GPS** + offline queue

8. Appendix — Agent fix prompt

Copy the block below into a new Agent chat when you want implementation. Do **not** implement full native background GPS (C2) in the first pass unless explicitly requested.

```
You are working in the React Native CLI app at:
/home/deep-patel/Application123/eb_geoemployee_tracker_mobile
```

Do NOT change business rules unless required to fix a bug.
Do NOT print or commit secrets (tokens, Maps keys, Firebase keys). Mask them.
Do NOT implement full native background GPS in the first pass unless I say so – that is a large follow-up (C2).
Prefer small, reviewable diffs. Match existing code style.

Context

- RN 0.86 / React 19, Redux Toolkit, Axios (15s default timeout), Odoo REST.
- Tokens in AsyncStorage. Single native stack (not auth-gated).
- Field tracking uses JS setInterval. Maps use WebView + company Google JS key from GET /field/map-config.

Fix in this pass (required)

C1 – Auth failure must leave the app
Files: App.tsx, AppNavigator.tsx, authSessionBridge
When refresh fails / onAuthFailure runs, reset navigation to Fetch URL (same as Settings logout). Do not leave the user on Home/tasks with a null token.
Keep a navigationRef and reset it from onAuthFailure (or subscribe to logout and reset).

C5 – Android icon
File: android/app/src/main/AndroidManifest.xml
Change android:icon from "@mipmap/ic_launcher" (space) to "@mipmap/ic_launcher".

C3 – Backup + tokens (minimal)

- Set android:allowBackup="false" on the application tag.
- Do not invent a full Keychain migration unless it is small and consistent; if you add encrypted storage, keep loadSession/saveSession/clearSession working for existing users.

C4 + H8 + TaskMap/TaskRouting/Location production logs
Remove or wrap in `if (\_\_DEV\_\_)`:

- locationTrackingService.ts coordinate logs
- apiClient.ts [TimeOff] console.logs
- TaskMapScreen / TaskRoutingScreen unguarded console.logs that include PII/coords/task names

Do not log tokens, passwords, or map API keys.

H9 – User-facing API errors
apiClient.ts extractErrorMessage + getAuthErrorMessage:
Never show raw Axios messages or unmapped backend strings. Map 401/403/404/409/422/429/5xx/timeout/network to existing AUTH_ERROR_MESSAGES (add keys if needed).

H2 – Time off double submit
TimeOffRequestScreen handleSubmit:
Lock immediately with a ref (and setSubmitting) BEFORE validation async work. Ignore extra taps. Keep existing validation and attachment behavior (leave created even if attach fails is OK, but do not create two leaves).

H1 – Time off list pagination
TimeOffCalendarScreen currently uses fetchHrLeaves default limit 20.
Load all pages or add load-more until the API has no more rows. Cancel with AbortController (already used). Do not break filters/chips.

H6 – Duplicate Time Off APIs
TimeOffCalendarScreen (focus) and TimeOffRequestScreen both fetch hr.leave.type and hr.leave.
Add a small in-memory cache (module-level or existing store) with short TTL so Request reuses Calendar data when navigating Calendar → Request. Still refetch after successful submit.

H5 – Task routing abort
TaskRoutingScreen loadTasks:
Use AbortController. handleRefresh must NOT use loadTasks() => false). Cancel on unmount and when chip/date/page changes.

H3 + H16 + H4 (Task Map) – do not over-refactor the whole 2000-line file
TaskMapScreen:

- Do not full-bootstrap (task-detail + map-config + rebuild HTML) on every focus. Bootstrap once per taskId; on later focus only resume GPS watch if needed.
- Pass AbortSignal into fetchFieldTaskDetail, fetchFieldMapConfig, fetchFieldDirections; abort on blur.
- Soften watchPosition: distanceFilter ~10, interval ~3000, fastestInterval ~2000 (keep cleanup on blur).
- Keep existing start/arrive/cancel/pause behavior.

H7 – Home resume duplicate task-detail
HomeScreen fetches task detail then opens TaskMap which fetches again.
Pass enough params so TaskMap can skip the first detail fetch if data is already present, OR a tiny session cache keyed by taskId with short TTL. Do not break resume → TaskCapture vs TaskMap routing.

H12 – Check-in timezone
CheckInScreen formatTimeLabel / formatTotalHours:
Stop blindly appending "Z". Display times consistently with how Odoo sends them (same approach as GeoHistoryDayScreen formatClock if that is correct).
Worked hours must not jump by timezone offset.

H18 – Get Help
HomeScreen: either hide Get Help or wire it to a real URL/screen. Do not leave onPress={() => {} }.

H21 / H17 – Client min tracking interval
locationTrackingService: raise MIN_TRACKING_INTERVAL_SECONDS to at least 10 (or 15). Do not allow 1s uploads.

M19 – Access gate too wide (if easy)
useAccessGateMonitor is enabled for every screen after splash, including Time Off.
Only enforce GPS+camera on field/camera screens (Home optional, CheckIn, DefaultRoute, TaskMap, TaskCapture, Selfie, ViewDetails, Geo). Do not block Time Off / Attendance / Settings / Profile on camera.

M24 – Time Off kebab
TimeOffCalendarScreen: hide the more-vert button until it has a real action (or implement cancel if the API already supports it – do not invent cancel API).

M23 – Silent session sync failure
attendanceStorage syncAttendanceFromServerNetwork: on network/API failure, keep local fallback but surface a toast once (or return a flag) so Home/CheckIn can show “Could not refresh attendance” without looping toasts.

C6 – Maps key (do not break maps)
Do not hardcode a Maps key. Keep fetching map-config.
Add a code comment that the key must be restricted in Google Cloud.
Do not log the key (TaskMap already logs keyLength – keep length only, and only in __DEV__).

Explicitly OUT OF SCOPE for this pass (unless leftover is tiny)

- C2 full background GPS / Android foreground service / iOS Always location (document a TODO in locationTrackingService only).
- H10 full offline point queue (optional small TODO).
- Removing unused npm packages (MapLibre, react-native-maps, nitro-image) – do it only if you have time at the end and the app still builds conceptually; do not break autolinking mid-fix.
- Rewriting apiClient.ts into many files.
- Changing Odoo backend.


```
## Verification
- Typecheck / lint the files you touch.
- Confirm AndroidManifest icon has no space.
- Confirm login still works: Fetch URL → Login → Home.
- Confirm 401 refresh failure would reset to Fetch URL (code path).
- Confirm Time Off submit cannot fire twice.
- Confirm Task Routing aborts in-flight GET on unmount.
- Confirm Task Map does not rebuild HTML on every blur/focus of the same task.

When done, list files changed and which bug IDs are fixed vs still open (especially C2, C3 encryption, H10).
```

9. Document control

Field	Value
Title	Geo Employee Tracker — Complete QA & Bug Audit Report
Version	1.0
Date	14 August 2026
Method	Static code review of the React Native CLI application
Code modified	None (audit only)
Related source	docs/QA_Audit_Report.md

End of report.